Lesson 5

Linux Command Line

Introduction

- A command-line user interface (CLI), also known as a console user interface and character user interface (CUI).
- A program which handles the interface is called a command language interpreter or **shell**.
- Command-line interfaces are often preferred by more advanced computer users, as they often provide a more concise and powerful means to control a program or operating system.
- Programs with command-line interfaces are generally easier to automate via scripting.
- Command-line interfaces include a number of programming languages such as Tcl/Tk, PHP and others, as well as utilities such as the compression utilities WinZip and UltimateZip, and some FTP and ssh/telnet clients.
- Pros:
  - Requires fewer resources, Concise access to options, Expert-friendly, Easier to automate via scripting or batch files, Commands can be logged to review or repeat, Easy to add special sub-options, Shorter to show multi-step actions
- Cons:
  - Requires help guide for commands, Commands can foster complex options, Not visually rich, results might scroll off-screen, Beginner-unfriendly

Introduction Cont'd…

- When you open a terminal emulator, by default you are in the home directory of the logged in user.
- You will see the name of the logged in user followed by the hostname.
- **$** means you are logged in as a regular user, whereas **#** means you are logged in as root.
- Unless you are performing administrative tasks or working inside root directories never work as root as it will change the permissions of all directories and files you worked on, making root the user of those directories and their content.
- The shell is a program that enables text based communication between the operating system and the user. It is usually a text mode program that reads the user's input and interprets it as commands to the system.
- There are several different shells on Linux, these are just a few:
  - Bourne-again shell (Bash), C shell (csh or tcsh, the enhanced csh), Korn shell (ksh), Z shell (zsh)
- On Linux the most common one is the Bash shell.

Introduction Cont’d…

- When using an interactive shell, the user inputs commands at a so-called prompt.
- For each Linux distribution, the default prompt may look a little different, but it usually follows this structure:
  - username@hostname current_directory shell_type
- the prompt for a regular user will likely look like this:
  - carol@mycomputer:~$
- The superuser’s prompt will look like this:
  - root@mycomputer:~#

Command Line Structure

- Most commands at the command line follow the same basic structure:
  - command [option(s)/parameter(s)...] [argument(s)...]
- Take the following command as an example:
  - $ ls -l /home
- Let's explain the purpose of each component:
- **Command**: Program that the user will run – ls in the above example.
- **Option(s)/Parameter(s)** : A "switch" that modifies the behavior of the command in some way, such as -l in the above example. Options can be accessed in a short and in a long form. For example, -l is identical to **--format=long.**
- Multiple options can be combined as well and for the short form, the letters can usually be typed together. For example, the following commands all do the same:
  - $ ls -al
  - $ ls -a -l
  - $ ls --all --format=long
- **Argument(**s): Additional data that is required by the program, like a filename or path, such as /home in the above example.
- The only mandatory part of this structure is the command itself. In general, all other elements are optional, but a program may require certain options, parameters or arguments to be specified.

Getting Help

- Each command has its own unique options, therefore documentation is key when working with a Linux system.
- Besides the **/usr/share/doc/** directory, which stores most of the documentation, various other tools provide information on using Linux commands.
- There are a multitude of methods to get help within the Linux command line.
- *man*, *help* and *info* are just a few of them. For Linux Essentials, we will be focusing on man and info as they are the most commonly used tools for obtaining help.

Getting Help Cont'd…

1. Built-in Help

- When started with the *--help* parameter, most commands display some brief instructions about their usage.
- Be aware that the instructions from --help often are rather brief compared to the other sources of documentation.

2. Man Pages

- Most commands provide a manual page or "man" page.
- This documentation is usually installed with the software and can be accessed with the man command.
- The command whose man page should be displayed is added to man as an argument:

  $ man mkdir

- This command opens the man page for *mkdir.*
- You can use the up and down arrow keys or the space bar to navigate through the man page. To exit the man page, press Q.

Getting Help Cont'd…

3. Info Pages

- Another tool that will help you while working with the Linux system are the *info* pages.
- The *info* pages are usually more detailed than the *man* pages and are formatted in hypertext, similar to web pages on the Internet.
- The *info* pages can be displayed like so:

  $ info mkdir

- For each *info* page, *info* reads an *info* file that is structured into individual nodes within a tree. Each node contains a simple topic and the *info* command contains hyperlinks that can help you move from one to the other.
- You can access the link by pressing enter while placing the cursor on one of the leading asterisks.
- Similar to *man*, the *info* tool also has page navigation commands. You can find out more about these command by pressing **?** while being on the info page.
- These tools will help you navigate the page easier as well as understand how to access the nodes and move within the node tree.

Revision Questions

1. Write a command that would explain to you the use of a command in Linux.
2. Differentiate between CLI and GUI, giving advantages and disadvantages of each.
3. Explain the use of *locate* command and give examples.